# Complete system — overview

Four functional sections • explicit interfaces • detailed circuits on sheets 2–16




Bundle lines contain separate conductors. Exact pins and nets are shown on detailed sheets.

POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

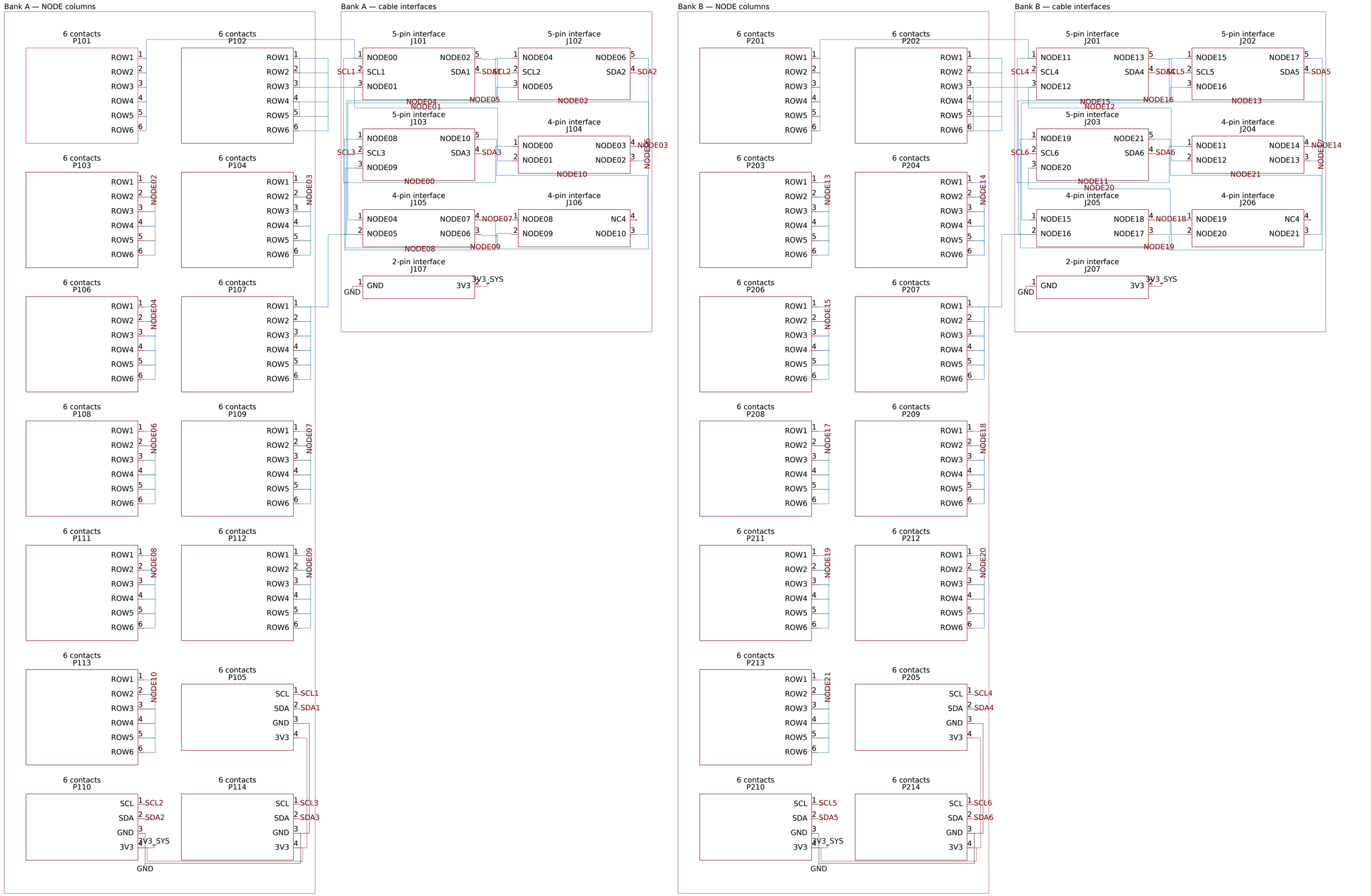
Revision8 candidate • circuit consistency checks; hardware qualification pending

1 / 16

A2 landscape

# Terminal contacts

Bank A and bank B • 22 independent NODEs • six OLED service interfaces




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

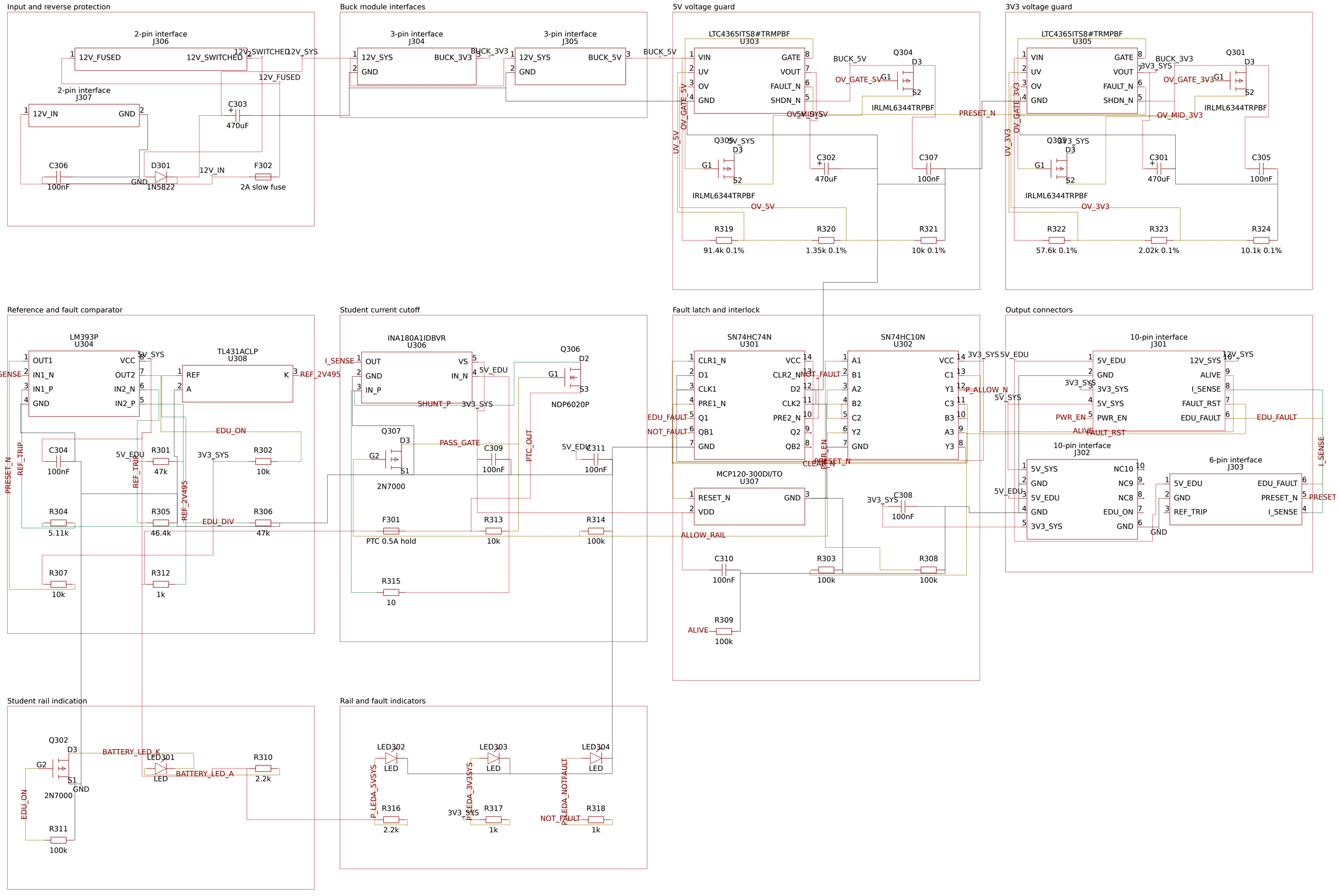
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Power board

Input → independent voltage guards → current trip → protected student output




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

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TedTronics — electrical schematic

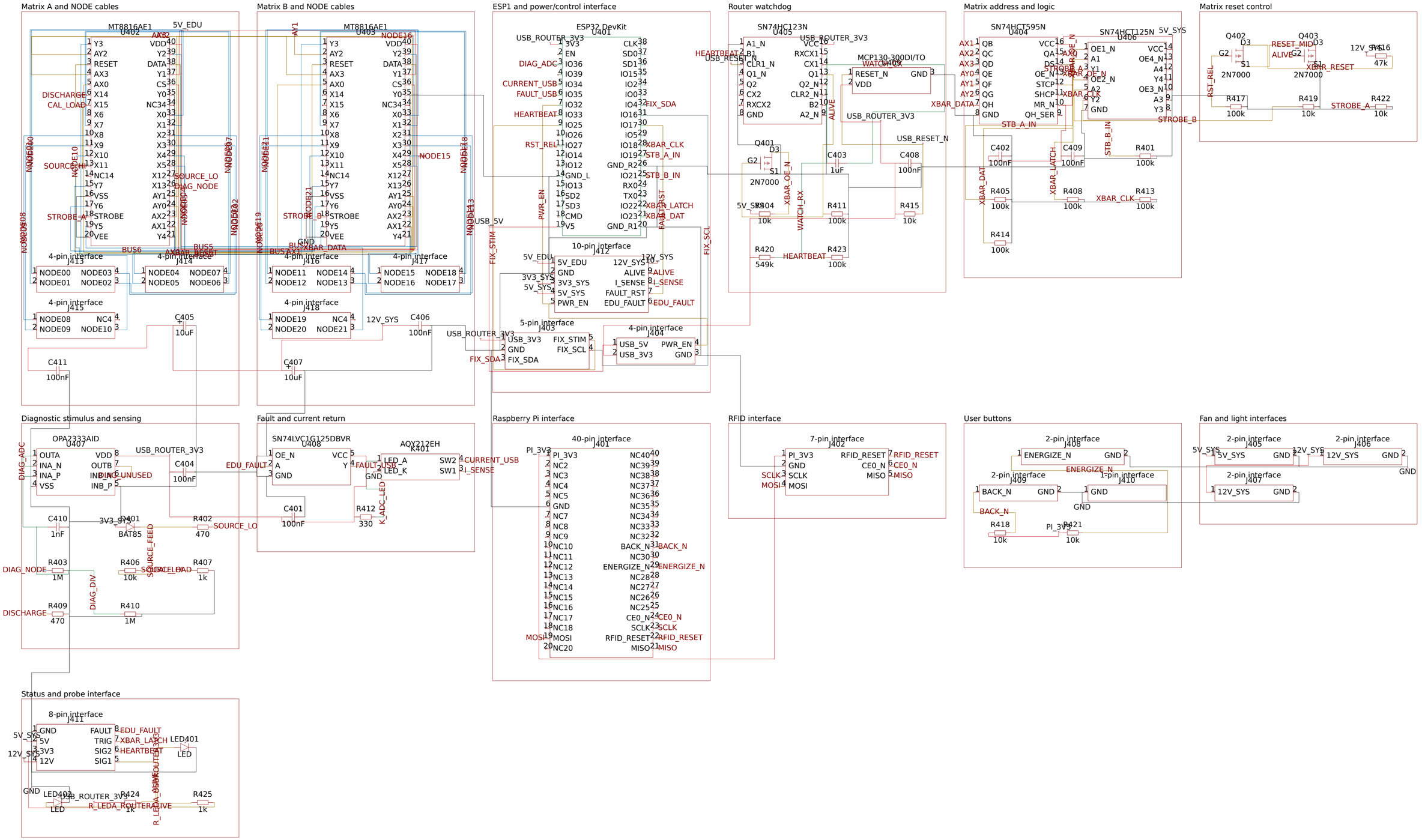
Revision8 candidate • circuit consistency checks; hardware qualification pending

3 / 16

A2 landscape

# Routing matrix and peripherals

ESP1 • watchdog • two MT8816 matrices • Pi and RFID interfaces




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

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TedTronics — electrical schematic

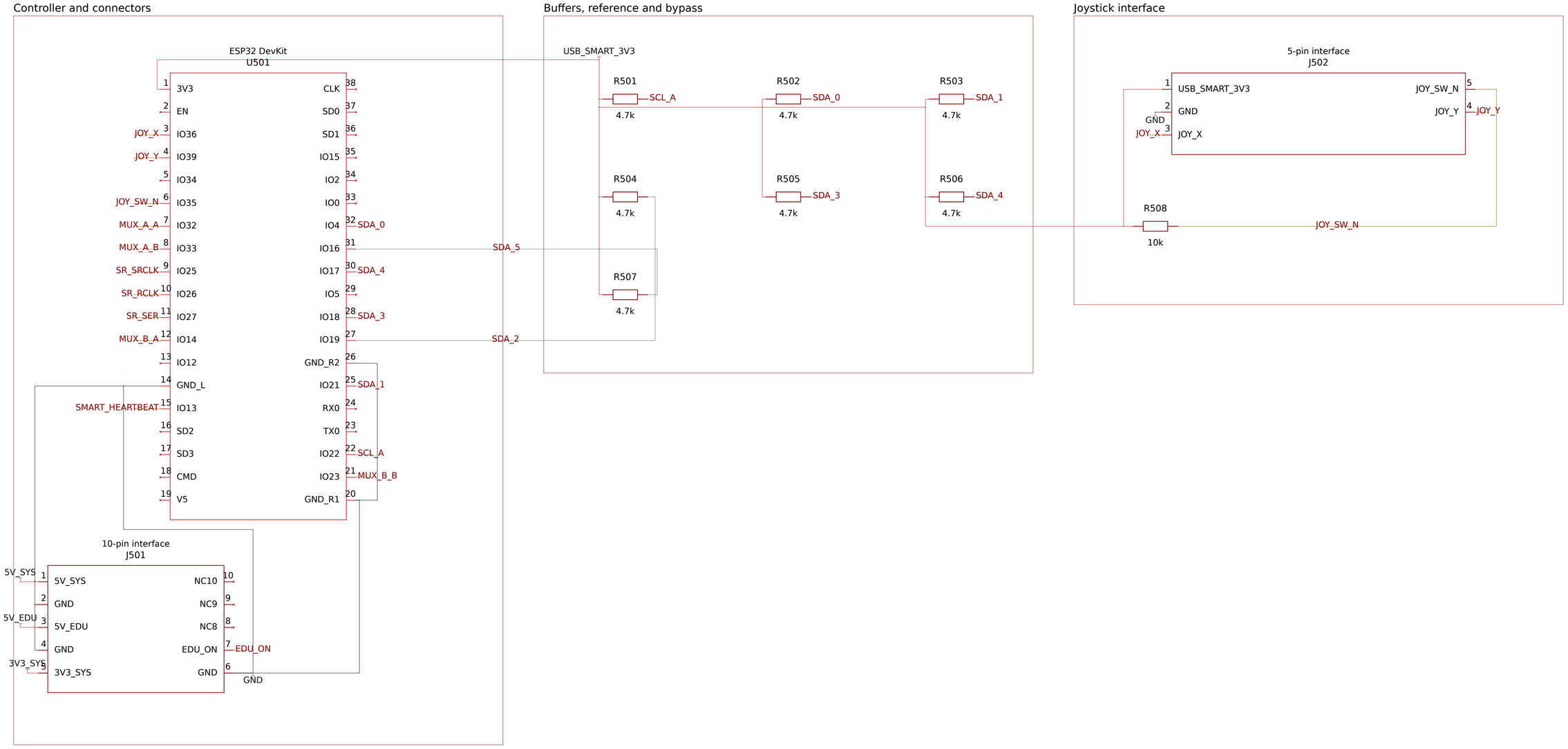
Revision8 candidate • circuit consistency checks; hardware qualification pending

4 / 16

A2 landscape

# Smart controller

ESP2 • six SDA channels • shared A-side SCL • bank selection • joystick




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

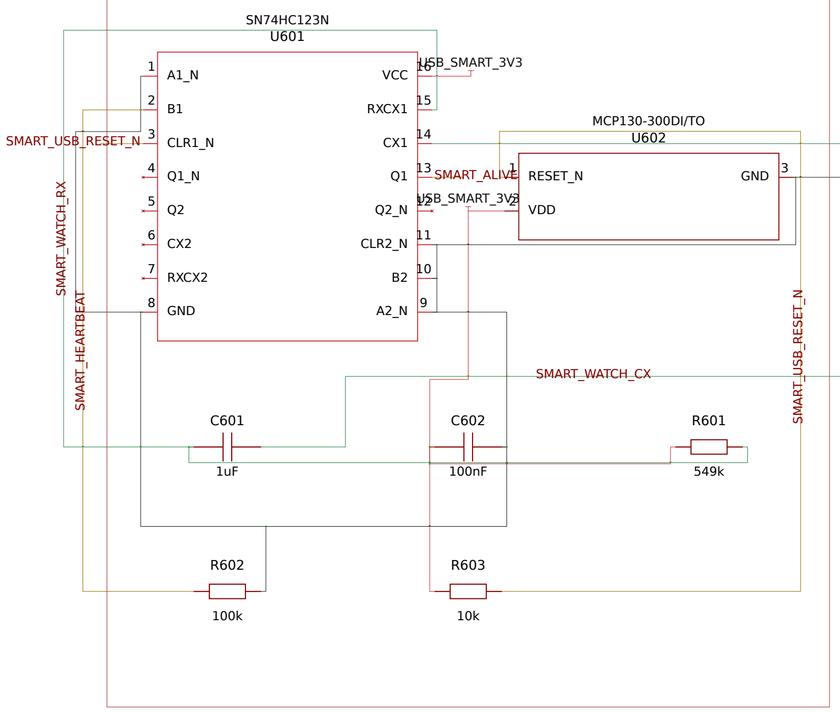
TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

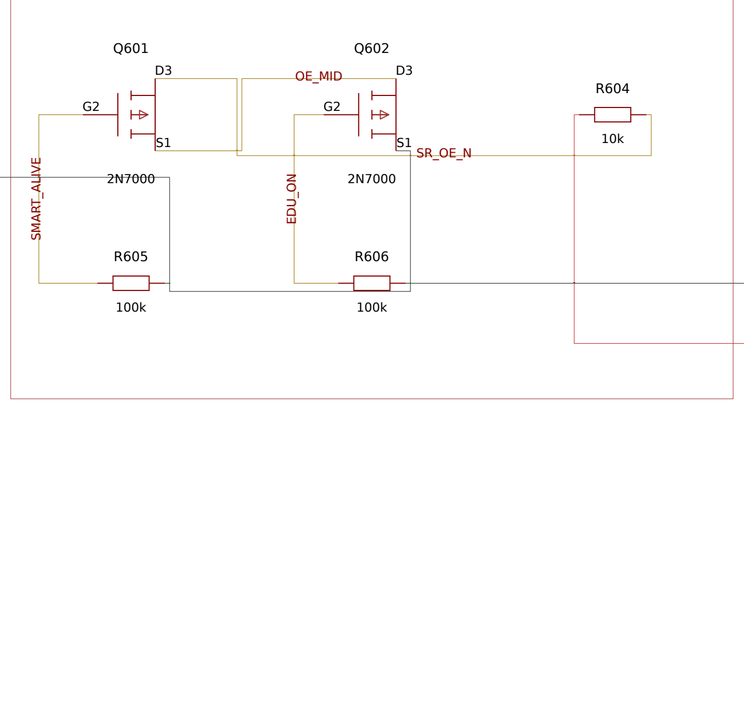
# Smart safety and register control

Independent watchdog • default-off output enable • three cascaded registers

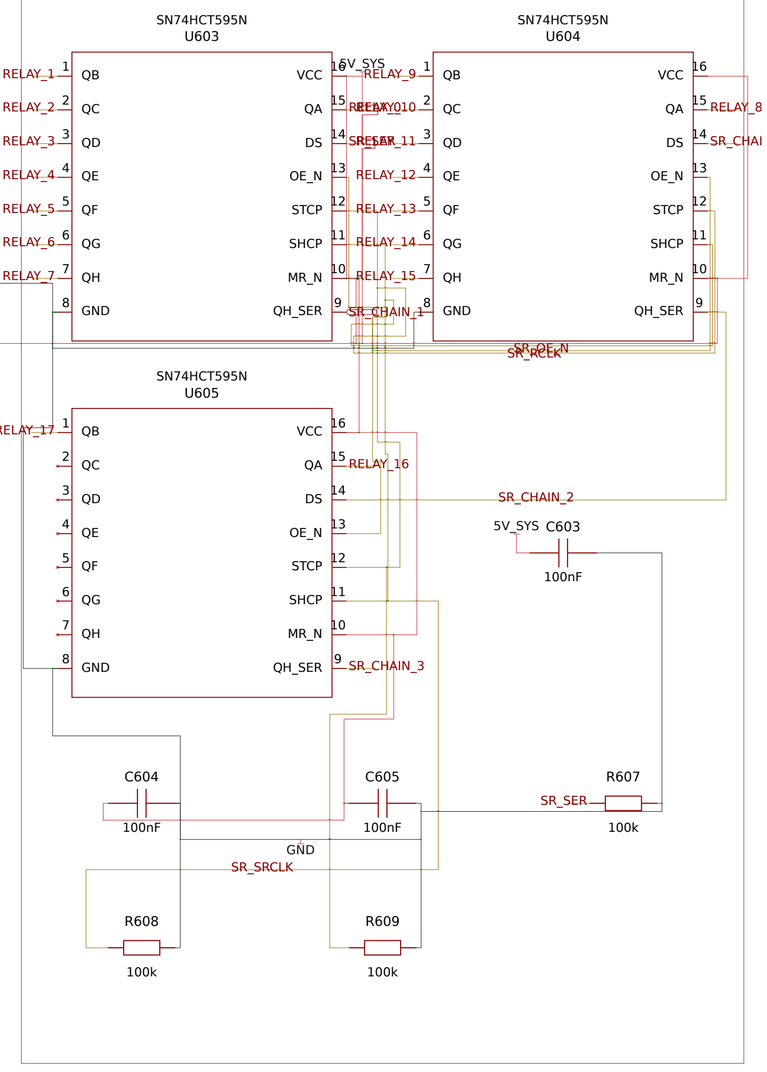
## Watchdog and reset




## Output enable and drivers




## Relay registers




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

6 / 16

A2 landscape

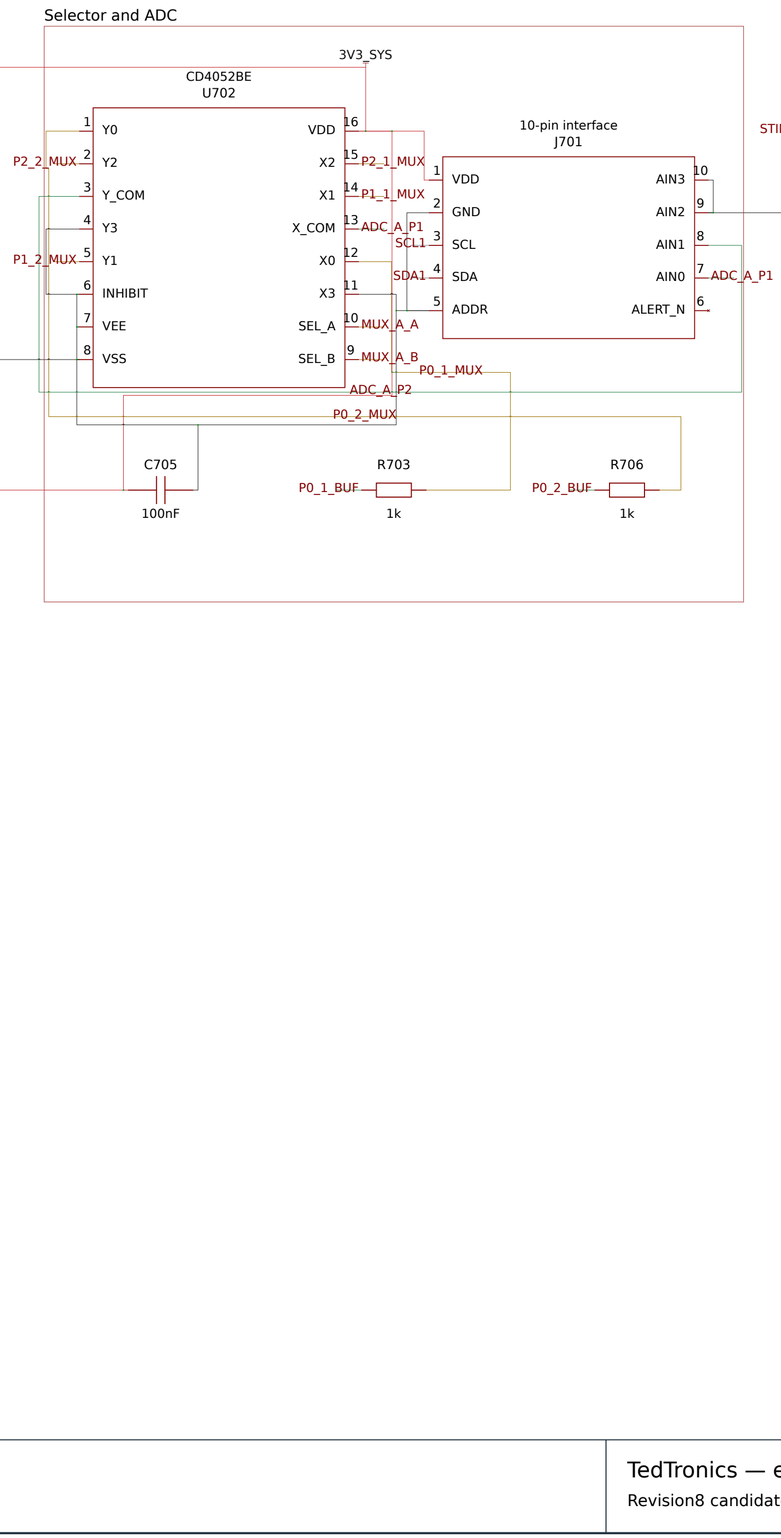
# Measurement bank A — Precision sensing and bank selector

Buffered voltage sensing • ADS1115 interfaces • named continuations to the smart positions

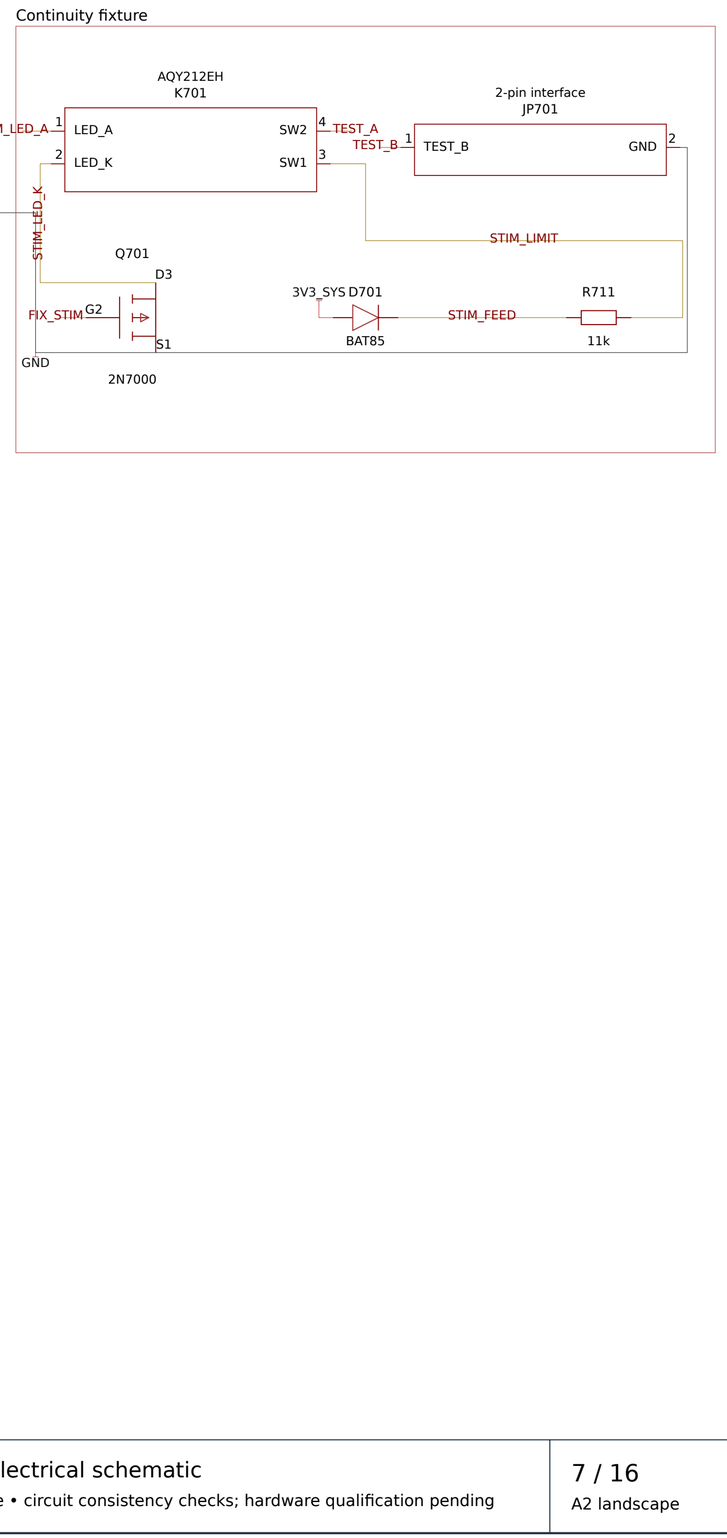
## Buffers, reference and bypass




## Selector and ADC



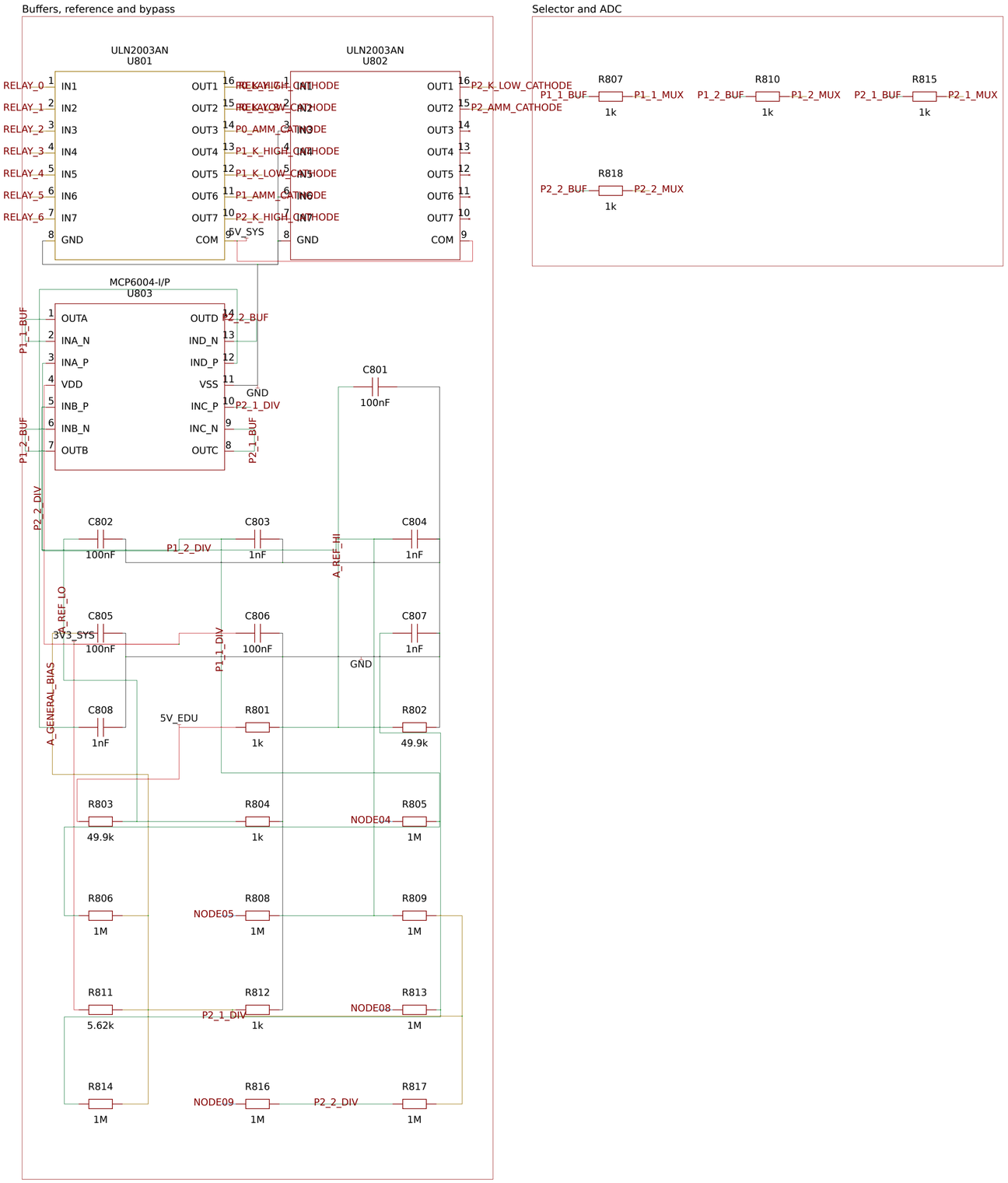

## Continuity fixture




## Controller and connectors

# Measurement bank A — Buffered inputs and references

Buffered voltage sensing • ADS1115 interfaces • named continuations to the smart positions




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

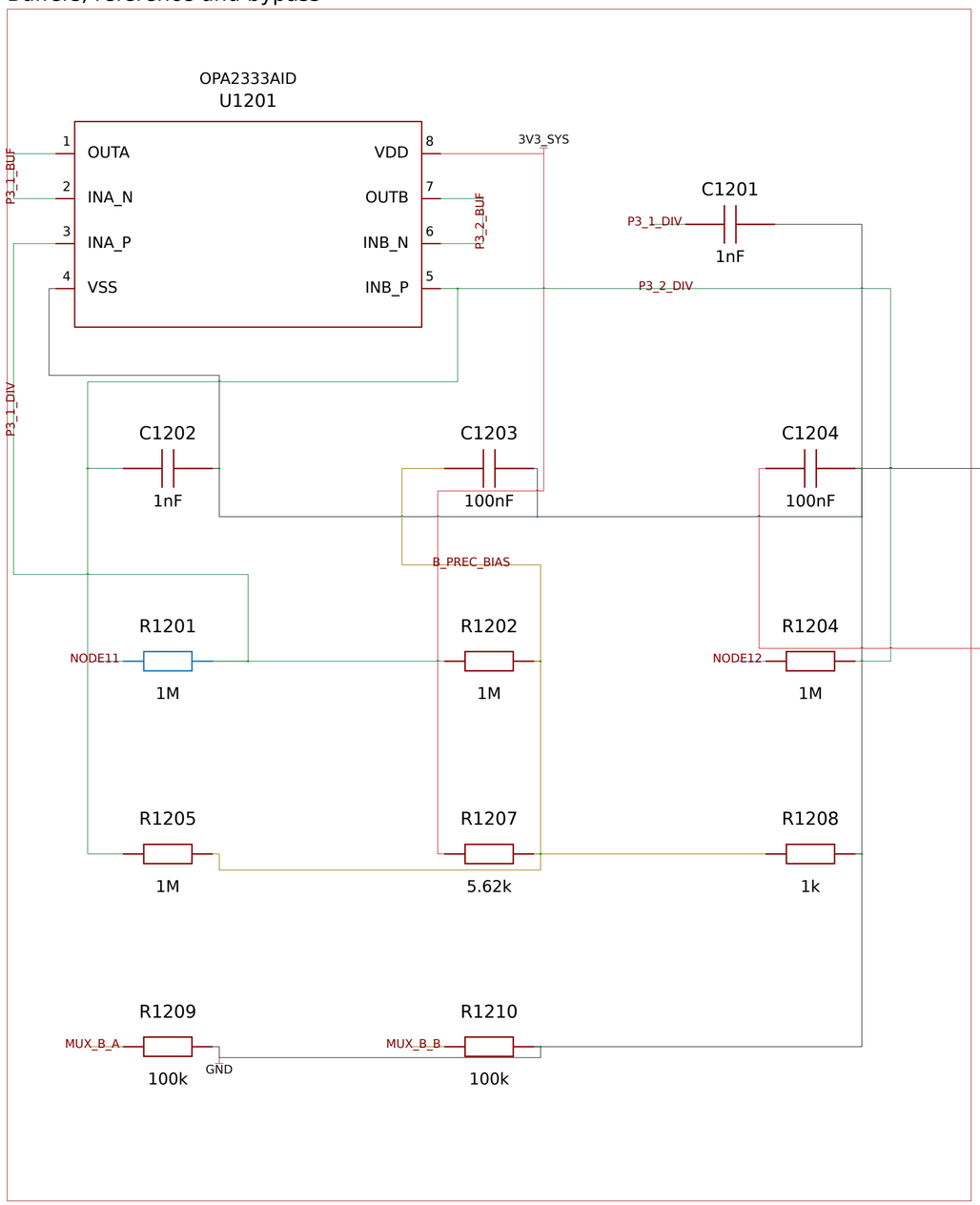
TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

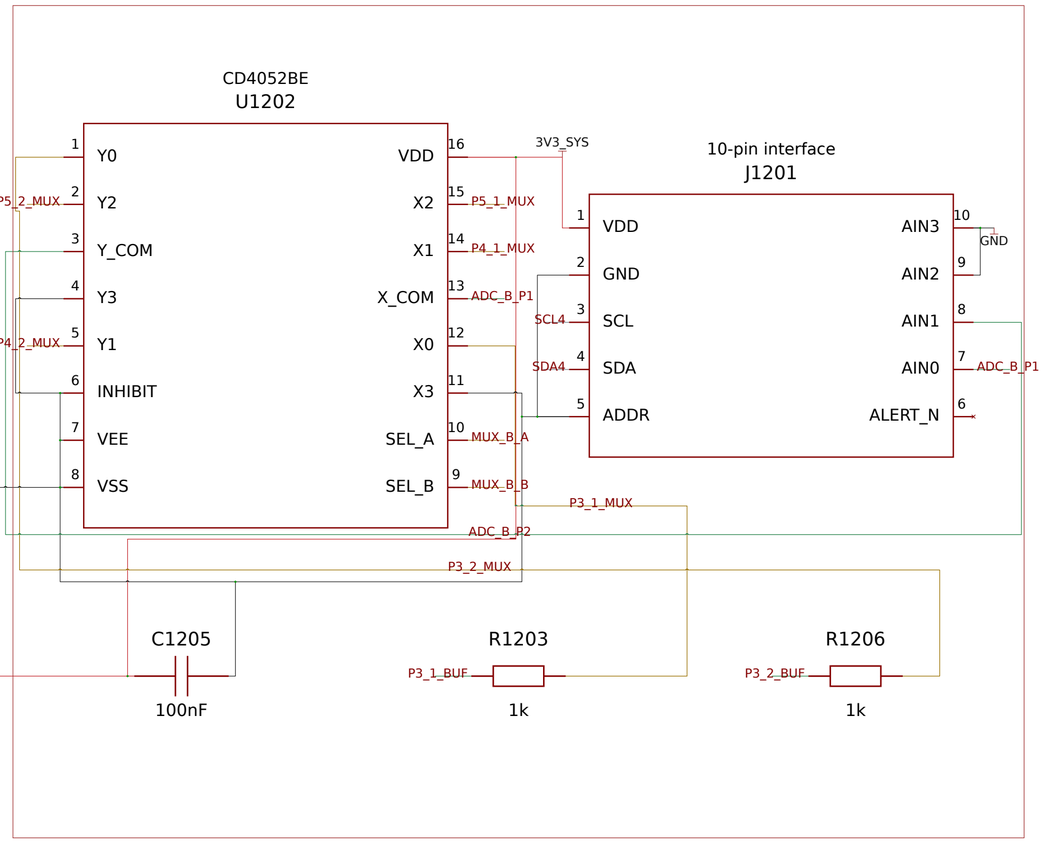
# Measurement bank B — Precision sensing and bank selector

Buffered voltage sensing • ADS1115 interfaces • named continuations to the smart positions

## Buffers, reference and bypass




## Selector and ADC




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

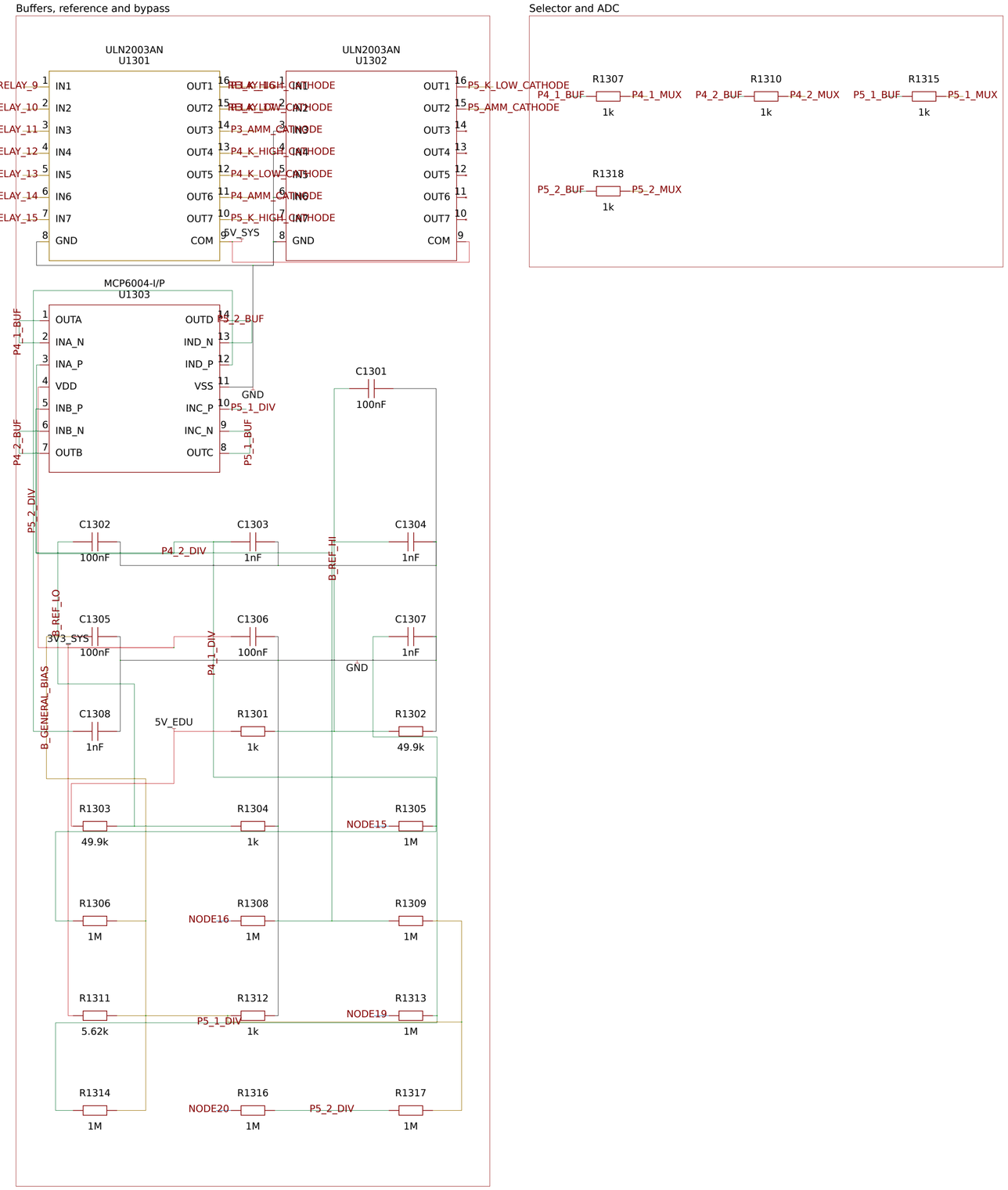
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Measurement bank B — Buffered inputs and references

Buffered voltage sensing • ADS1115 interfaces • named continuations to the smart positions




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

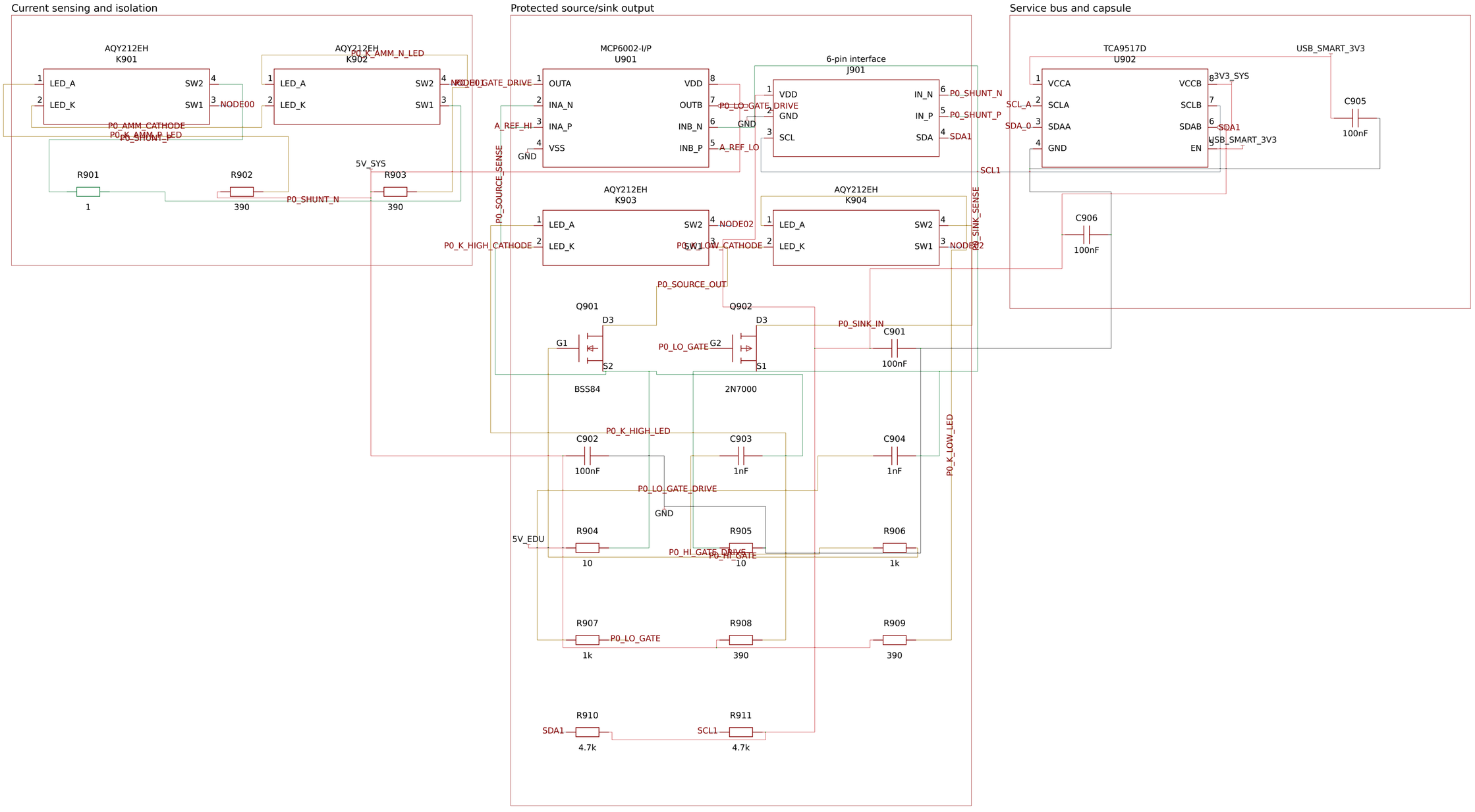
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Smart position 1 — bank A

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

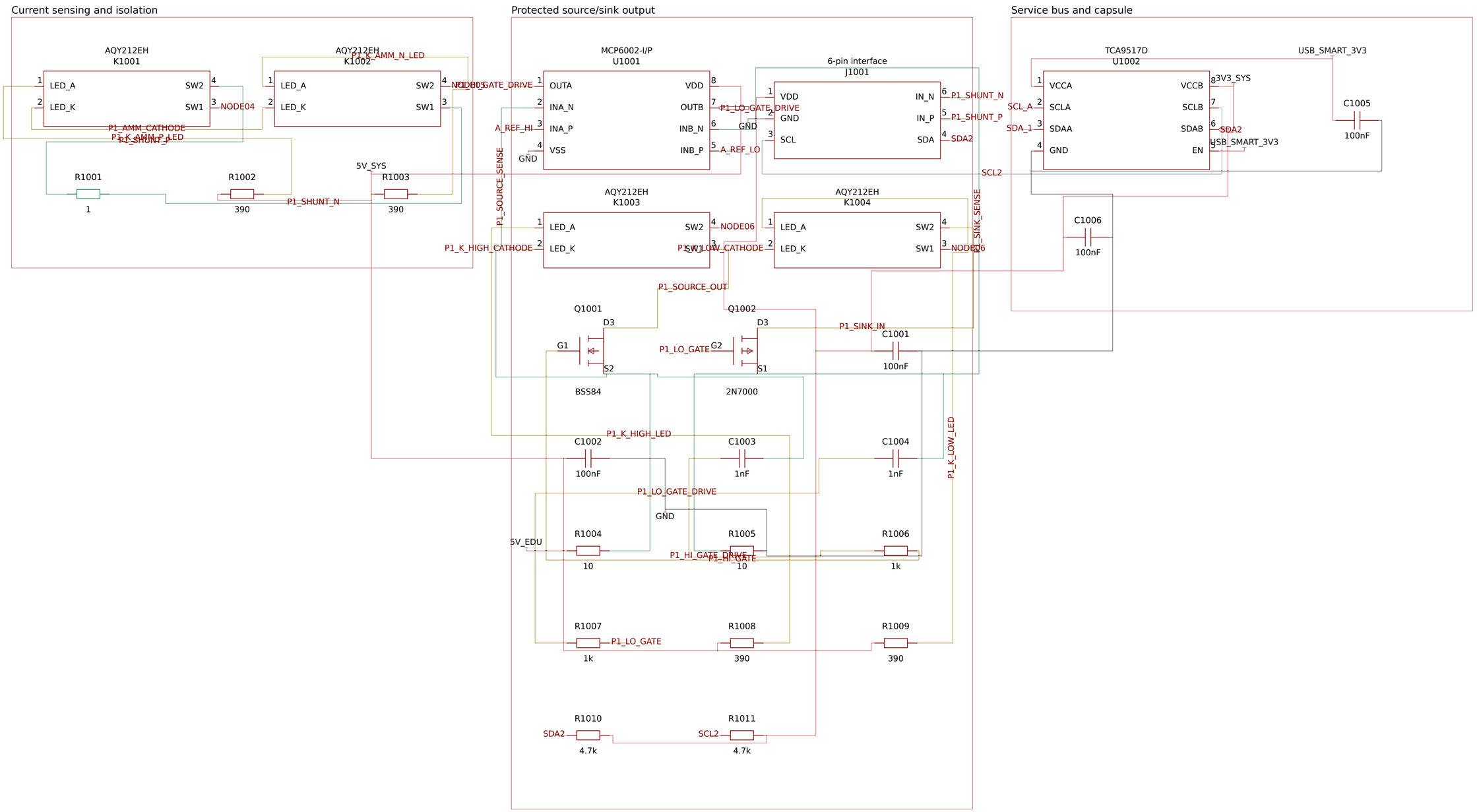
Revision8 candidate • circuit consistency checks; hardware qualification pending

11 / 16

A2 landscape

# Smart position 2 — bank A

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

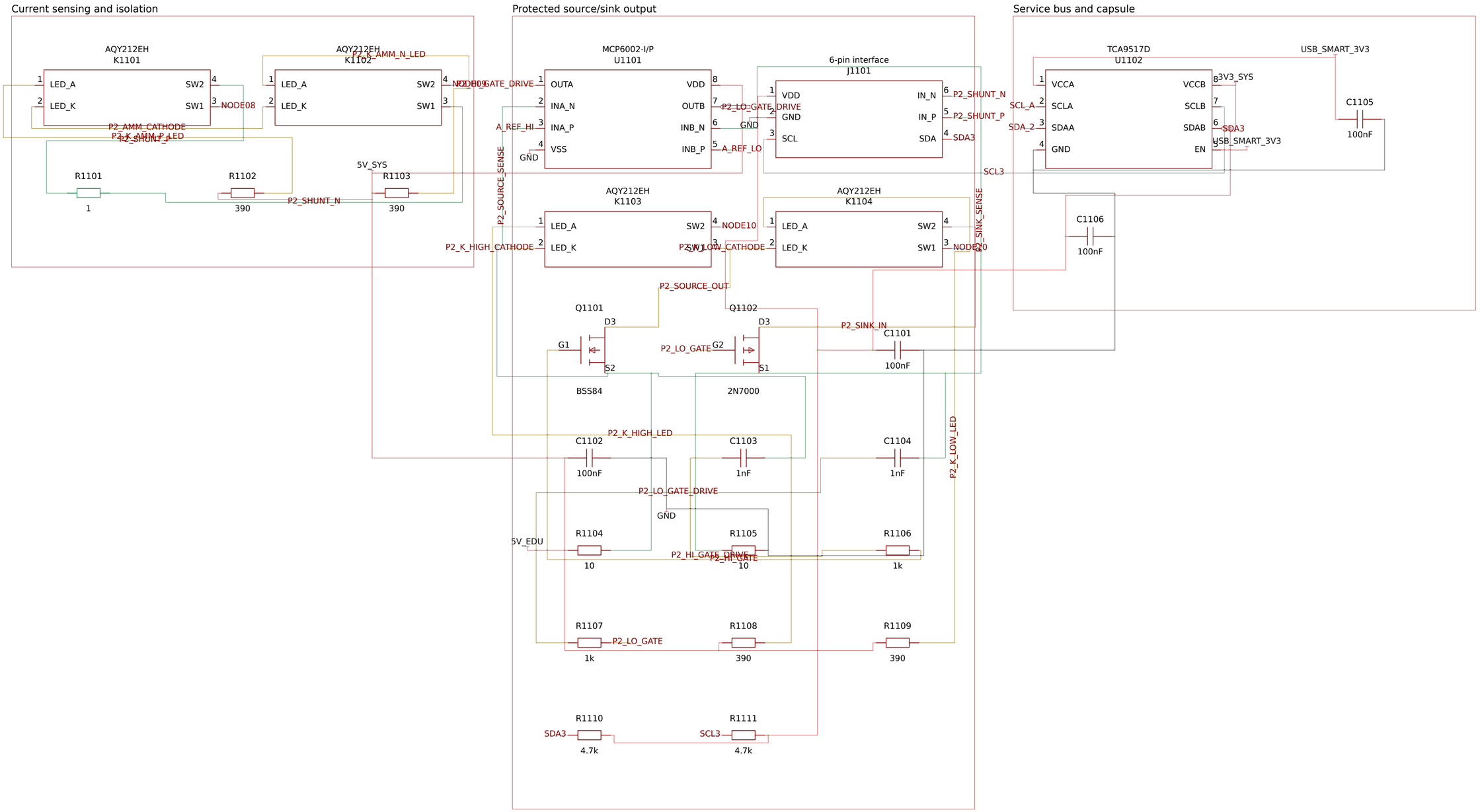
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Smart position 3 — bank A

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

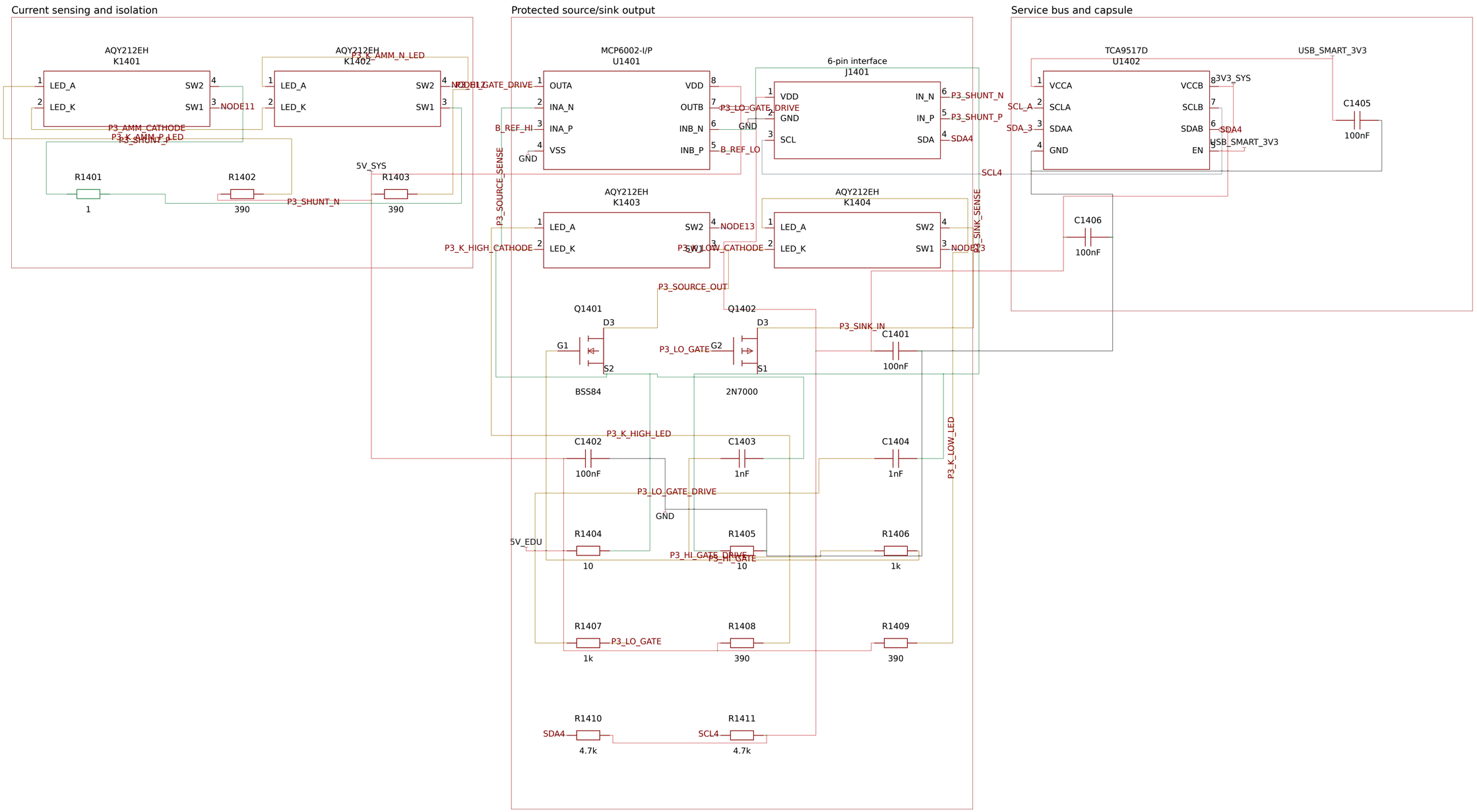
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Smart position 4 — bank B

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

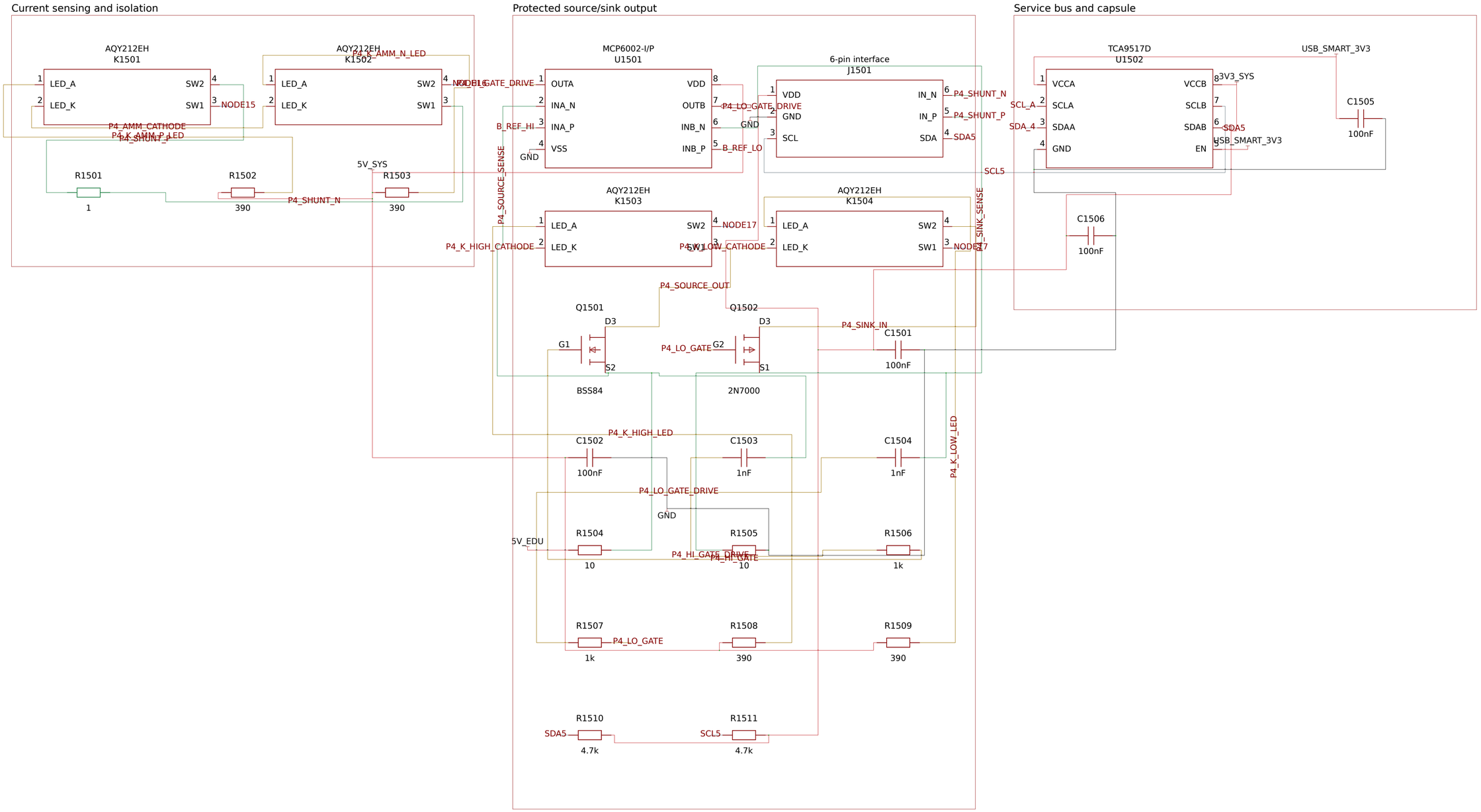
Revision8 candidate • circuit consistency checks; hardware qualification pending



A2 landscape

# Smart position 5 — bank B

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

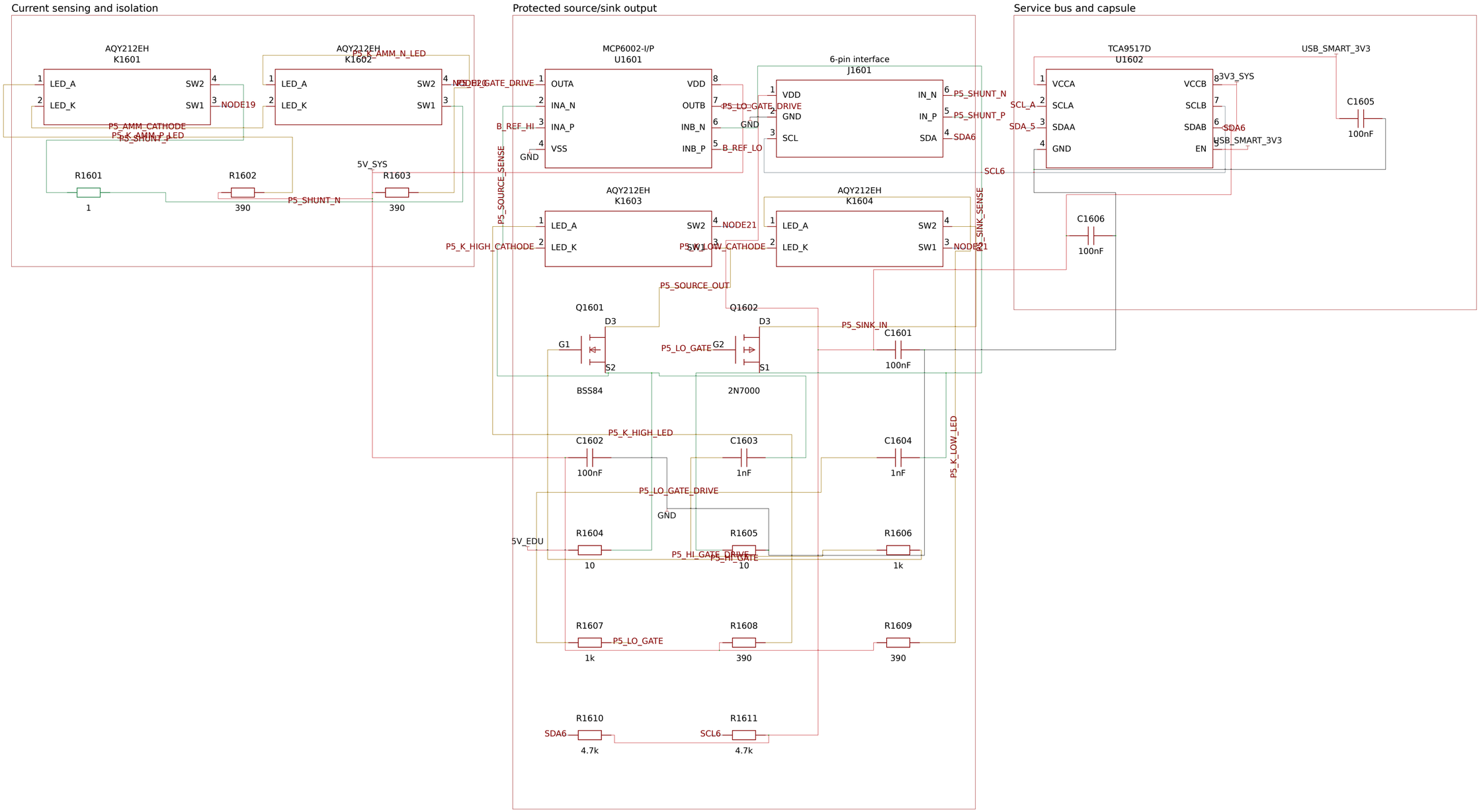
Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

# Smart position 6 — bank B

Isolated service bus • protected P3 source/sink • isolated current path • OLED capsule interface




POWER red • GND black • NODE blue • ANALOG green • CONTROL ochre • I²C grey/white

Junction dot = connected. Named pins and flags identify continuations; see net index. Modules show interface pins.

TedTronics — electrical schematic

Revision8 candidate • circuit consistency checks; hardware qualification pending

16 / 16

A2 landscape